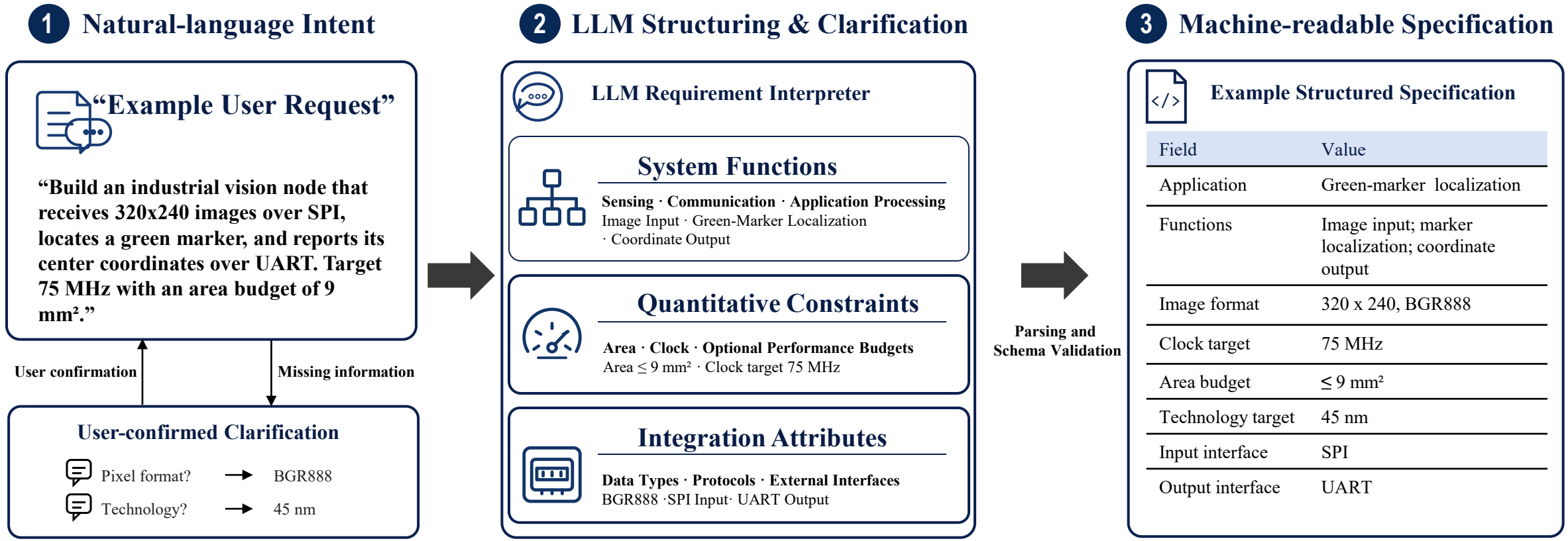

1 Natural-language Intent
"Example User Request"
"Build an industrial vision node that receives 320x240 images over SPI, locates a green marker, and reports its center coordinates over UART. Target 75 MHz with an area budget of 9 mm²."
User confirmation
Missing information
User-confirmed Clarification
Pixel format?
BGR888
Technology?
45 nm
2 LLM Structuring & Clarification
LLM Requirement Interpreter
System Functions
Sensing · Communication · Application Processing
Image Input · Green-Marker Localization
· Coordinate Output
Quantitative Constraints
Area · Clock · Optional Performance Budgets
Area ≤ 9 mm² · Clock target 75 MHz
Integration Attributes
Data Types · Protocols · External Interfaces
BGR888 ·SPI Input· UART Output
Parsing and
Schema Validation
3 Machine-readable Specification
Example Structured Specification
Field
Value
Application
Green-marker localization
Functions
Image input; marker localization; coordinate output
Image format
320 x 240, BGR888
Clock target
75 MHz
Area budget
≤ 9 mm²
Technology target
45 nm
Input interface
SPI
Output interface
UART